

A Bayesian framework with adaptive elastic nets for the inference of Gaussian graphical models

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Abstract

Estimating conditional independence graphs from high-dimensional Gaussian data is challenging because methods must detect relevant edges while rigorously controlling statistical errors. We propose a Bayesian framework based on a prior accounts for degree heterogeneity edge sparsity, and graph topology the graph. The resulting posterior distribution is incorporated into a multiple testing procedure for graph inference with false discovery rate control. Computation is carried out through a combination of adaptive elastic nets and a variational expectation–maximization algorithm. In simulations, the method achieves reliable false discovery rate control while maintaining strong power, especially in heterogeneous networks such as graphs with hubs, and remains competitive under structural misspecification. Applications to breast cancer gene expression data and financial return networks show that the method yields sparse and interpretable conditional dependence graphs while retaining the most stable interactions detected by competing approaches. **Keywords:** Gaussian graphical models; Bayesian inference; Elastic net; Spike-and-slab prior; Stochastic block model.

1 Introduction

1.1 Problem statement

Understanding dependence structure among variables is a central problem in modern statistical analysis. In many scientific applications, the aim is not merely to quantify marginal association, but to identify the pattern of direct interactions governing the joint behaviour of a multivariate system. Graphical representations provide a natural and interpretable summary of such structure, with variables represented by nodes and relationships by edges. Gaussian graphical models (Lauritzen, 1996; Koller and Friedman, 2009) provide a standard framework for learning such structure from multivariate Gaussian data. Owing to their interpretability and flexibility, they have been widely used in genomics, ecology, neuroscience and finance (Shutta et al., 2022; Ranciati et al., 2021; Da Fermo et al., 2024; Diebold and Yilmaz, 2014; Smith et al., 2011; Varoquaux et al., 2010).

In high-dimensional settings, where the number of variables exceeds the sample size, substantial work has been devoted to inference in Gaussian graphical models; see, for example, Meinshausen and Bühlmann (2006), Friedman et al. (2008), Liu (2013), Gan et al. (2019) and Kilian et al. (2024). Despite these advances, it remains difficult to control statistical errors while detecting as many true edges as possible. This issue is important in practice, because false discoveries may lead to incorrect scientific conclusions and costly empirical validation. In this paper we consider control of the false discovery rate (FDR) (Benjamini and Hochberg, 1995), defined as the expected proportion of false discoveries among selected edges. Our aim is to develop a method for edge detection that controls the FDR while retaining high power.

Our approach is developed in a Bayesian framework, in which estimation yields posterior probabilities of edge inclusion. Such quantities have already been used for graph recovery, for example by Gan et al. (2019), who proposed the BAGUS method and recovered the graph by thresholding posterior probabilities at 0.5. Building on ideas from multiple testing, we use posterior edge probabilities to construct a procedure that controls the FDR at a user-specified nominal level. To improve power, we account for structural heterogeneity in graph topology, motivated by the fact that networks arising in applications often exhibit hubs, community structure and varying connectivity patterns.

1.2 Related work

In Gaussian graphical models, the primary object of interest is conditional independence as marginal correlations may reflect indirect associations induced by other variables. Two variables are conditionally independent if and only if the corresponding entry of the precision matrix, defined as the inverse covariance matrix, is zero. The conditional independence graph is therefore determined by the non-zero off-diagonal entries of the precision matrix.

A major class of methods for inferring conditional independence graphs in high-dimensional Gaussian graphical models is based on penalized likelihood estimation of the precision matrix K ,

where the penalty induces sparsity. These methods can be written in the form

$$\hat{K} = \arg \max_{K \succ 0} \{\ell(K; S) - \text{pen}(K)\},$$

where $\ell(K; S) = n(\log \det K - \text{tr}(SK))/2$ up to an additive constant, the Gaussian loglikelihood based on the sample covariance matrix S , and $\text{pen}(K)$ is a penalty function that encourages sparsity in the graph structure. A prominent example is the graphical lasso (Yuan and Lin, 2007; Banerjee et al., 2008; Friedman et al., 2008), which uses an element-wise ℓ_1 penalty,

$$\text{pen}(K) = \lambda \|K\|_1, \quad \|K\|_1 = \sum_{i < j} |K_{ij}|.$$

This penalty shrinks some coefficients to zero and leads to computationally efficient procedures, but it treats all entries of K equally, regardless of their structural role in the network. Such uniform penalization may be suboptimal when the underlying graph is heterogeneous.

To address this issue, Ambroise et al. (2009) proposed SIMoNe, which introduces latent groups labels $Z_i \in \{1, \dots, Q\}$ for each node i and uses a block-weighted ℓ_1 penalty of the form $\text{pen}(K) = \|\lambda_Z(K)\|_1$. Here, $\lambda_Z(K)$ denotes a group-dependent weighting of the entries of K , so that the amount of penalization applied to K_{ij} depends on the groups to which nodes i and j belong, thereby allowing different degrees of penalization within and between groups. This encourages heterogeneous sparsity patterns adapted to block structure. However, the method does not provide posterior probabilities for edges and so does not yield a mechanism for false discovery rate control.

An alternative is to work in a Bayesian framework, where sparsity is induced through a prior distribution on the entries of K rather than through an explicit penalty. Wang (2012) showed that the graphical lasso estimator is the maximum a posteriori estimator under independent Laplace priors on the off-diagonal entries of K . The BAGUS method of Gan et al. (2019) is based on a hierarchical Bayesian model in which the distribution of each entry K_{ij} depends on whether the corresponding edge is present. It relies on a spike-and-slab prior formed by two Laplace distributions with different scales. The corresponding maximum a posteriori estimator can be viewed as arising from a non-convex penalty, $\text{pen}(K) = \sum_{i < j} \text{pen}_{\text{SS}}(K_{ij})$, where $\text{pen}_{\text{SS}}(K_{ij}) = -\log \pi(K_{ij})$ and $\pi(K_{ij})$ denotes the spike-and-slab prior. Computation is made tractable through an expectation-maximization algorithm that solves a sequence of weighted graphical lasso problems. Unlike many competing approaches, BAGUS separates estimation of the precision matrix from inference on the graph: K is first estimated, and the graph is then recovered by thresholding approximate posterior edge probabilities at $1/2$.

For the purposes of this paper, BAGUS has two limitations. First, unlike SIMoNe, it does not exploit heterogeneity in graph topology. Second, thresholding posterior probabilities at $1/2$ provides no guarantee on the false discovery rate.

Only a limited number of procedures explicitly control the false discovery rate in Gaussian graphical model selection. One of the earliest is the graphical false discovery control procedure of Liu (2013), based on multiple testing of conditional correlations. More recently, knockoff-based methods have been extended to Gaussian graphical models (Li and Maathuis, 2021), following the general framework of Barber and Candès (2015). These approaches provide finite-sample false discovery rate guarantees

by constructing synthetic variables that preserve the dependence structure while enabling valid edge selection. Another recent contribution is due to Kilian et al. (2024), who adapted the noisy stochastic block model framework of Rebafka et al. (2022) to the Gaussian graphical model setting and used graph-based test statistics to control the false discovery rate. These approaches also have limitations. Testing procedures such as that of Liu (2013) may suffer from low power in high dimensions. Knockoff methods, although attractive theoretically, require the construction of valid knockoffs and are mainly developed for moderate-dimensional settings. The NSBM-based approach depends strongly on the quality of the input statistics, and its performance may deteriorate under model misspecification.

1.3 Contributions

We propose a Bayesian procedure for inferring conditional independence graphs in heterogeneous high-dimensional Gaussian graphical models under false discovery rate control. In the sequel, we refer to the proposed procedure as HSS-GGM, short for heterogeneous spike-and-slab Gaussian graphical model. Our main contributions are as follows.

1. We introduce a Bayesian model for conditional independence graph estimation that combines a stochastic block model for heterogeneous graph structure with a spike-and-slab prior on the precision matrix. This yields an edge-specific adaptive elastic-net procedure, in which the amount of shrinkage depends on posterior edge probabilities.
2. We develop a multiple testing procedure for graph selection that uses posterior edge probabilities to control the false discovery rate.
3. We develop a computationally tractable inference algorithm that alternates between variational estimation of the latent block structure and model parameters, and adaptive elastic-net estimation of the precision matrix.
4. Numerical experiments on simulated data show that the proposed method controls the false discovery rate at nominal levels across different graph topologies while maintaining high power and remaining robust to model misspecification. Applications to breast cancer gene expression data and financial return networks show that the method yields sparse and interpretable conditional independence graphs while retaining stable interactions detected by competing methods.

2 Bayesian formulation for Gaussian graphical models

2.1 Statistical model and notation

We consider the problem of recovering the conditional independence graph of a Gaussian random vector

$$X = (X_1, \dots, X_p)^T \in \mathbb{R}^p, \quad X \sim \mathcal{N}(0, \Sigma).$$

Given n independent copies $X^{(1)}, \dots, X^{(n)}$, we write

$$\mathbf{X} = (X_{i,j})_{1 \leq i \leq n, 1 \leq j \leq p} \in \mathbb{R}^{n \times p}$$

for the data matrix, whose i th row is $(X^{(i)})^\top$. Hence X_j is the j th component of X , $X_j^{(i)}$ is the j th component of the i th observation, and $X_{i,j} = X_j^{(i)}$. Let $K = \Sigma^{-1}$ denote the corresponding precision matrix. In a Gaussian graphical model, conditional independence is equivalent to a zero off-diagonal entry of K , that is,

$$X_i \perp\!\!\!\perp X_j \mid \mathbf{X}_{-\{i,j\}} \iff K_{ij} = 0. \quad (1)$$

We denote the support of K by the binary matrix $A = (A_{ij})$, defined for $i \neq j$ by

$$A_{ij} = \mathbb{1}_{\{K_{ij} \neq 0\}}, \quad A_{ii} = 0.$$

Thus A is the adjacency matrix of the conditional independence graph associated with K where nodes correspond to variables and edges encode conditional dependencies between them. Recovering this graph may be viewed as a multiple testing problem, in which each potential edge corresponds to the hypotheses

$$H_{0,ij} : A_{ij} = 0 \quad \text{versus} \quad H_{1,ij} : A_{ij} = 1, \quad (2)$$

for all $(i, j) \in \mathcal{A}$, where $\mathcal{A} = \{(i, j) : 1 \leq i < j \leq p\}$. A Bayesian hierarchical model with prior distributions on the graph A and the precision matrix K provides a natural framework for computing posterior probabilities of edge inclusion and for deriving multiple testing rules from these posterior quantities.

2.2 Hierarchical spike-and-slab prior on the precision matrix

We adopt a Bayesian approach to test the hypotheses in (2), using a prior that allows larger values of K_{ij} when an edge is present and stronger concentration around zero when no edge is present. This leads naturally to a spike-and-slab formulation.

Spike-and-slab priors are a Bayesian tool for inducing sparsity. They were introduced by Mitchell and Beauchamp (1988) as a mixture of a point mass at zero, called the spike, and a diffuse distribution for non-zero coefficients, called the slab. Although conceptually appealing, such discrete formulations generally lead to difficult posterior computation. Continuous spike-and-slab priors address this issue by replacing the Dirac spike with a sharply concentrated continuous distribution. A notable example is the spike-and-slab lasso of Ročková (2018); Ročková and George (2018, 2016), which uses two Laplace components with different scales and yields tractable penalized likelihood estimators in sequence and regression models. In the context of Gaussian graphical models, Gan et al. (2019) adopted a spike-and-slab prior formed by two Laplace distributions for estimation of the precision matrix. While such a formulation preserves computational tractability, a Laplace slab may induce substantial shrinkage of large signals.

To alleviate this effect, we replace the Laplace slab with a Gaussian slab. The Laplace spike preserves strong concentration around zero, whereas the Gaussian slab has a softer shrinkage effect

on large coefficients and is therefore better suited to retaining strong edge signals. More precisely, for each off-diagonal entry K_{ij} , $i < j$, we define

$$K_{ij} \mid A_{ij} \sim \begin{cases} \text{Laplace}(\xi_0), & \text{if } A_{ij} = 0 \quad (\text{spike}), \\ \mathcal{N}(0, \sigma_1^2), & \text{if } A_{ij} = 1 \quad (\text{slab}), \end{cases} \quad (3)$$

where the Laplace distribution with rate parameter $\xi_0 > 0$ has density

$$g_{\xi_0}(x) = \frac{\xi_0}{2} \exp(-\xi_0|x|), \quad x \in \mathbb{R},$$

and the Gaussian density with mean 0 and variance σ_1^2 is

$$\phi(x; 0, \sigma_1^2) = \frac{1}{\sqrt{2\pi} \sigma_1} \exp\left(-\frac{x^2}{2\sigma_1^2}\right), \quad x \in \mathbb{R}.$$

For the diagonal entries, which do not correspond to edges, we follow Deshpande et al. (2017) and assign independent exponential priors

$$K_{ii} \sim \text{Exponential}(\sigma_1), \quad i = 1, \dots, p. \quad (4)$$

The quantities ξ_0 and σ_1 are hyperparameters whose calibration is discussed in Section ???. We also impose the positive-definiteness constraint $K \succ 0$. This completes the specification of the conditional prior on $K \mid A$. It remains to specify a prior distribution for the graph A .

2.3 Structured prior on the conditional independence graph

A simple choice is to draw the edge indicators A_{ij} independently from a Bernoulli(η) distribution with common success probability η , as is assumed in many sparse graphical model priors (Gan et al., 2019; Vogels et al., 2024; Xi and Ruffieux, 2024). However, such a homogeneous prior does not capture the heterogeneous connectivity patterns often observed in real networks, where some nodes act as hubs, others remain peripheral, and connection probabilities vary across the graph.

To account for this heterogeneity, we model the graph A through a stochastic block model (Holland et al., 1983; Nowicki and Snijders, 2001). Each node is assigned to one of Q latent groups according to

$$Z_i \sim \text{Multinomial}(1, \pi), \quad i = 1, \dots, p, \quad (5)$$

where $\pi = (\pi_1, \dots, \pi_Q)$ satisfies $\pi_q > 0$ and $\sum_{q=1}^Q \pi_q = 1$. Conditionally on the labels $Z = (Z_1, \dots, Z_p)$, the edge indicators are independent and satisfy

$$A_{ij} \mid (Z_i = q, Z_j = \ell) \sim \text{Bernoulli}(\omega_{q\ell}), \quad (i, j) \in \mathcal{A}, \quad (6)$$

where $\omega \in (0, 1)^{Q \times Q}$ is symmetric and $\omega_{q\ell}$ denotes the connection probability between groups q and ℓ . We denote by $\theta = (\pi, w)$ the unknown model parameters arising from the stochastic block model.

Combining (3)–(6) gives the joint hierarchical prior

$$\Pi(K, A, Z) = \left[\prod_{i=1}^p \Pi(Z_i) \right] \left[\prod_{i < j} \Pi(A_{ij} \mid Z_i, Z_j) \right] \left[\prod_{i < j} \Pi(K_{ij} \mid A_{ij}) \right] \left[\prod_{i=1}^p \Pi(K_{ii}) \right] \mathbb{1}_{\{K > 0\}}. \quad (7)$$

3 Posterior approximation and inference

3.1 Inference strategy by alternating optimization

Our primary inferential goal is graph recovery, that is, deciding whether an edge is present between two nodes i and j . Under the proposed hierarchical model, the natural Bayesian target is the posterior edge probabilities

$$\text{pr}(A_{ij} = 1 \mid \mathbf{X}), \quad 1 \leq i < j \leq p,$$

which are obtained by integrating out the latent precision entries $K_{i,j}$ and the latent block memberships Z_{ij}

$$\text{pr}(A_{ij} = 1 \mid \mathbf{X}) = \sum_{q,\ell} \int \text{pr}(A_{ij} = 1 \mid K_{ij}, Z_i = q, Z_j = \ell) \Pi(K_{ij}, Z_i = q, Z_j = \ell \mid \mathbf{X}) dK_{ij}.$$

Obviously, these quantities are difficult to estimate. Instead, our procedure aims to estimate the conditional posterior probabilities

$$\text{pr}(A_{ij} = 0 \mid K_{ij}, Z_i, Z_j; \theta),$$

which admit a closed-form expression and can be evaluated once K , the latent blocks Z and the model parameters θ have been estimated. Note that in the hierarchical model, the latent graph A and the observations $\mathbf{X}$ are conditionally independent given K , which implies that the posterior distribution of A given $(\mathbf{X}, K)$ does not depend on $\mathbf{X}$, that is $\Pi(A \mid \mathbf{X}, K) = \Pi(A \mid K)$. To jointly estimate all unknown quantities in the model, we propose an alternating optimization algorithm in which each set of unknown quantities is updated while keeping the others fixed. The algorithm consists of two steps. In the first step, when K is fixed, the model corresponds to a so-called noisy stochastic block model, and a variational expectation–maximization (VEM) algorithm can be used to fit the model to the data (Rebafka et al., 2022). This step relies on a variational approximation of the conditional distribution of (A, Z) given K . The algorithm provides estimates of the latent block memberships Z in addition to parameter estimates. The second step updates the precision matrix K by maximizing a penalized likelihood criterion, where the penalty takes the form of a weighted elastic net induced by our spike-and-slab prior. The ℓ_1 component, induced by the Laplace spike, promotes sparsity by shrinking coefficients toward zero, while the ℓ_2 component, induced by the Gaussian slab, allows for larger values. The penalty is edge-specific, with weights given by the posterior edge probabilities, resulting in adaptive shrinkage across edges. The two steps of the inference algorithm are described in more detail in the following two subsections.

3.2 Variational expectation–maximization

The general VEM algorithm for inference on (A, Z) and the parameters θ , for a fixed precision matrix K , was first proposed in Rebafka et al. (2022). It is adapted here to the specific choice of Gaussian and Laplace distributions for the precision matrix entries. Globally, the VEM algorithm maximizes a lower bound on the marginal log-likelihood $\log \Pi(\mathbf{X}, K; \theta)$, known as the evidence lower bound,

$$\mathcal{J}(\mathbf{X}, K, Q; \theta) = \mathbb{E}_Q[\log \Pi(\mathbf{X}, K, A, Z; \theta)] + \mathcal{H}(Q),$$

where $Q(A, Z)$ denotes a variational distribution approximating the conditional distribution of the latent variables (A, Z) given K under the current value of θ , and $\mathcal{H}(Q)$ denotes its entropy. The algorithm alternates between two steps until convergence. The M-step maximizes $\mathcal{J}$ with respect to θ while keeping Q fixed. The E-step maximizes $\mathcal{J}$ with respect to Q while keeping θ fixed, which is equivalent to minimizing the Kullback–Leibler divergence

$$\text{KL}(Q(A, Z) \parallel \Pi(A, Z \mid X, K; \theta)),$$

thereby yielding the best approximation to the conditional distribution of the latent variables. A common choice for the variational distribution is the mean-field approximation (Jordan et al., 1999; Daudin et al., 2008; Blei et al., 2017), using the class of factorized distributions of the form

$$Q(A, Z) = \Pi(A \mid K, Z; \theta) \prod_{i=1}^p \tau_{i, Z_i}, \quad (8)$$

where $\tau_{iq} \in [0, 1]$ and $\sum_{q=1}^Q \tau_{iq} = 1$. The parameter τ_{iq} represents the posterior probability that node i belongs to block q . The variational E-step provides updates of all variational parameters $\tau = (\tau_{iq})_{1 \leq i \leq p, 1 \leq q \leq Q}$, by solving a fixed-point equation, as well as estimates of the posterior edge probabilities

$$\rho_{q\ell}^{i,j} := \text{pr}(A_{ij} = 1 \mid K_{ij}, Z_i = q, Z_j = \ell; \theta) = \frac{\omega_{q\ell} \phi(K_{ij}; 0, \sigma_1^2)}{\omega_{q\ell} \phi(K_{ij}; 0, \sigma_1^2) + (1 - \omega_{q\ell}) g_{\xi_0}(K_{ij})}, \quad (9)$$

for each node pair (i, j) and block pair (q, ℓ) , with $\phi(\cdot; 0, \sigma_1^2)$ denoting the Gaussian density with mean 0 and variance σ_1^2 , and $g_{\xi_0}(\cdot)$ denoting the Laplace density with rate parameter ξ_0 . More details are provided in Appendix ??.

3.3 Precision matrix estimation by adaptive elastic-net regression

The precision matrix K is updated by maximizing the expected complete-data log-likelihood under the current variational distribution

$$\mathbb{E}_Q[\log \Pi(\mathbf{X}, K, A, Z; \theta)]. \quad (10)$$

This expectation effectively approximates the unknown latent variables using their estimated posterior distributions. Maximizing (10) with respect to K amounts to maximizing

$$\ell(K; S) - \sum_{i < j} \sum_{q, \ell} \tau_{iq} \tau_{j\ell} \left[\rho_{q\ell}^{i,j} \frac{K_{ij}^2}{2\sigma_1^2} + (1 - \rho_{q\ell}^{i,j}) \xi_0 |K_{ij}| \right] - \sigma_1 \sum_{i=1}^p K_{ii}, \quad (11)$$

where $\ell(K; S) = \frac{n}{2}(\log \det K - \text{tr}(SK))$ is, up to an additive constant, the Gaussian log-likelihood based on the sample covariance matrix S . As direct maximization is involved, we follow Meinshausen and Bühlmann (2006) and Chiquet et al. (2011) and adopt a pseudo-likelihood approach that splits the optimization problem into p independent regression problems. To state the pseudo likelihood, for each $i \in \{1, \dots, p\}$, we denote $\mathbf{X}_i \in \mathbb{R}^n$ the i th column of the data matrix $\mathbf{X}$ and $\mathbf{X}_{-i} \in \mathbb{R}^{n \times (p-1)}$ the matrix obtained by removing that column. Then the conditional distribution of $\mathbf{X}_i$ given the other variables is Gaussian with

$$\mathbf{X}_i | \mathbf{X}_{-i} \sim \mathcal{N}(\mathbf{X}_{-i}\beta_i, \sigma_i^2 I_n), \quad \sigma_i^2 = \frac{1}{K_{ii}}, \quad \beta_i = -\frac{K_{-i,i}}{K_{ii}} \in \mathbb{R}^{p-1},$$

where $K_{-i,i} \in \mathbb{R}^{p-1}$ denotes the i th column of K with its diagonal entry removed. We introduce the pseudo log-likelihood

$$\check{\ell}(K; S) = \sum_{i=1}^p \log p(\mathbf{X}_i | \mathbf{X}_{-i}, K) = \sum_{i=1}^p \left[-\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma_i^2 - \frac{1}{2\sigma_i^2} \|\mathbf{X}_i - \mathbf{X}_{-i}\beta_i\|_2^2 \right]. \quad (12)$$

Finally, in (11), we replace $\ell(K; S)$ with $\check{\ell}(K; S)$ leading to an optimization problem that amounts to solving p separate regression problems of dimension $p - 1$. More precisely, to update the i th column of K we solve

$$\hat{\beta}_i = \arg \min_{\beta \in \mathbb{R}^{p-1}} \left\{ \frac{1}{2\sigma_i^2} \|\mathbf{X}_i - \mathbf{X}_{-i}\beta\|_2^2 + \sum_{j \neq i} P_{1,ij} |\beta_{ij}| + \sum_{j \neq i} P_{2,ij} \beta_{ij}^2 \right\}, \quad (13)$$

with weights

$$P_{1,ij} = \xi_0(1 - p_{ij})|K_{ii}|, \quad P_{2,ij} = \frac{1}{2\sigma_1^2} p_{ij} K_{ii}^2, \quad p_{ij} = \sum_{q, \ell} \tau_{iq} \tau_{j\ell} \rho_{q\ell}^{i,j}. \quad (14)$$

The problem in (13) corresponds to a weighted elastic-net regression, where the variational posterior edge probabilities p_{ij} modulate the weighting of the Lasso and Ridge penalties. This induces an adaptive trade-off between sparsity and smooth shrinkage. In particular, edges with high posterior probability are mainly governed by ridge-type penalization, whereas edges with low posterior probability are mainly driven by Lasso-type shrinkage. The criterion in (13) is convex and becomes strictly convex as soon as $P_{2,ij} > 0$ for all $j \neq i$. Therefore, each nodewise subproblem is well-posed and admits a unique minimizer whenever $p_{ij} > 0$ for all $j \neq i$.

Input: Data matrix $\mathbf{X}$, variational parameters (τ, ρ) , hyperparameters (σ_1, ξ_0) , current precision matrix K

Output: Updated precision matrix K

for each node $i = 1, \dots, p$ **do**

 Compute the variational posterior probability of edge presence p_{ij} for $j \neq i$ with (??);

 Compute the edge-penalty weights $P_{1,ij}$ and $P_{2,ij}$ for $j \neq i$ with (14);

 Solve the weighted elastic-net problem (13) to obtain $\hat{\beta}_i \in \mathbb{R}^{p-1}$;

 Update the i -th column of K with (15);

Symmetrise the matrix K using the OR rule (16)

Algorithm 1: Adaptive elastic-net estimation of the precision matrix

Input: Data matrix $\mathbf{X}$, number of blocks Q , hyperparameters (ξ_0, σ_1)

Output: Estimates of K, τ, ρ, θ

Initialization;

Initialize $K^{(0)}, \tau^{(0)}, \theta^{(0)}$ and $t = 0$

Compute $\rho^{(0)}$ from $K^{(0)}$ and $\theta^{(0)}$ with equation (9)

while not converged **do**

 Set $t = t + 1$;

 Update the precision matrix $K^{(t)}$ using Algorithm 1 from variational parameter $(\tau^{(t-1)}, \rho^{(t-1)})$ and current $K^{(t-1)}$;

 Update the variational parameters $\tau^{(t)}, \rho^{(t)}$ and model parameter $\theta^{(t)}$ using the VEM algorithm detailed in Appendix ?? from $K^{(t)}$ and current $(\tau^{(t-1)}, \rho^{(t-1)}, \theta^{(t-1)})$

Algorithm 2: Alternating inference algorithm

Once $\hat{\beta}_i$ has been obtained, the corresponding entries of the precision matrix are updated as

$$\hat{K}_{ii} = \frac{1}{\hat{\sigma}_i^2} = \frac{n}{\|\mathbf{X}_i - \mathbf{X}_{-i}\hat{\beta}_i\|_2^2}, \quad \hat{K}_{ji} = -\hat{\beta}_{ij} \hat{K}_{ii}, \quad j \neq i. \quad (15)$$

Since the nodewise procedure generally yields a non-symmetric matrix, we symmetrize $\hat{K}$ using the OR rule: for each pair (i, j) , we retain the coefficient with the largest absolute value, together with its sign, namely

$$\hat{K}_{ij}^{\text{sym}} = \hat{K}_{ji}^{\text{sym}} = \begin{cases} \hat{K}_{ij}, & \text{if } |\hat{K}_{ij}| \geq |\hat{K}_{ji}|, \\ \hat{K}_{ji}, & \text{otherwise.} \end{cases} \quad (16)$$

As our primary objective is graph recovery rather than precision matrix estimation, global positive-definiteness of $\hat{K}$ is not enforced at this stage.

Algorithm 1 summarizes the precision matrix update, while Algorithm 2 describes the full inference algorithm alternating between VEM updates and precision matrix updates.

3.4 Graph selection by multiple testing

We now address graph inference through the multiple testing problem given in (2), where our goal is to decide whether an edge is present. We propose a multiple testing procedure based on the ℓ -values introduced by Efron et al. (2001); Efron (2004) and defined as the conditional posterior probabilities

$$\ell_{ij}(K, Z, \theta) = \text{pr}(A_{ij} = 0 \mid K_{ij}, Z_i, Z_j; \theta), \quad (i, j) \in \mathcal{A}. \quad (17)$$

In the spike-and-slab model, this quantity has the explicit form

$$\ell_{ij}(K, Z, \theta) = 1 - \rho_{Z_i Z_j}^{ij} = \frac{(1 - \omega_{Z_i Z_j}) g_{\xi_0}(K_{ij})}{(1 - \omega_{Z_i Z_j}) g_{\xi_0}(K_{ij}) + \omega_{Z_i Z_j} \phi(K_{ij}; 0, \sigma_1^2)}. \quad (18)$$

Intuitively, small values of ℓ_{ij} provide evidence against $H_{0,ij}$, and therefore in favour of the presence of an edge. A multiple testing procedure consists in thresholding the ℓ -values at an appropriate level t chosen to control the FDR defined as

$$\text{FDR}_\theta(t) = \mathbb{E}_\theta \left[\frac{\sum_{(i,j) \in \mathcal{A}} (1 - A_{ij}) \mathbb{1}_{\{\ell_{ij}(K, Z, \theta) \leq t\}}}{\sum_{(i,j) \in \mathcal{A}} \mathbb{1}_{\{\ell_{ij}(K, Z, \theta) \leq t\}} \vee 1} \right]. \quad (19)$$

. Here $\mathbb{E}_\theta$ denotes expectation under the model indexed by θ .

In multiple testing literature, the marginal false discovery rate (MFDR) is often used as a convenient substitute for the FDR. The MFDR is defined as

$$\text{MFDR}_\theta(t) = \frac{\mathbb{E}_\theta \left[\sum_{(i,j) \in \mathcal{A}} (1 - A_{ij}) \mathbb{1}_{\{\ell_{ij}(K, Z, \theta) \leq t\}} \right]}{\mathbb{E}_\theta \left[\sum_{(i,j) \in \mathcal{A}} \mathbb{1}_{\{\ell_{ij}(K, Z, \theta) \leq t\}} \right]}, \quad (20)$$

with the convention $0/0 = 0$. We reject the $H_{0,ij}$ whenever $\ell_{ij} \leq t_\alpha$ with t_α such that $\text{MFDR}_\theta(t_\alpha) = \alpha$. To circumvent the explicit calculation of this threshold, we use the following quantities

$$q_{ij}(K, Z, \theta) = \text{MFDR}_\theta(\ell_{ij}(K, Z, \theta)), \quad (i, j) \in \mathcal{A}, \quad (21)$$

These quantities, called q -values, were used in Rebafka et al. (2022) and can be interpreted as adjusted ℓ -values in the sense that they can be compared directly with the target FDR level α .

The q -values admit an explicit expression in our model, as detailed in Appendix ??.

Using the estimates of K , Z , and θ obtained from Algorithm 2, we define the empirical q -values as

$$\widehat{q}_{ij} = q_{ij}(\widehat{K}, \widehat{Z}; \widehat{\theta}), \quad (i, j) \in \mathcal{A}, \quad (22)$$

where the node clustering $\widehat{Z}$ is obtained from the variational parameters through

$$\widehat{Z}_i = \arg \max_{1 \leq q \leq Q} \widehat{\tau}_{iq}, \quad i = 1, \dots, p. \quad (23)$$

Then, the final estimated graph at nominal level α is

$$\widehat{A}_{ij}^\alpha = \mathbb{1}_{\{\widehat{q}_{ij} \leq \alpha\}}, \quad (i, j) \in \mathcal{A}. \quad (24)$$

3.5 Calibration of hyperparameters and selection of the number of blocks

The spike-and-slab hyperparameters (ξ_0, σ_1) and the number of blocks Q are calibrated jointly using an appropriate Bayesian information criterion. The choice of (ξ_0, σ_1) is guided by the theoretical bounds established in Section 4.1. In particular, the theory requires the spike level ξ_0 to be of order $\sqrt{n \log p}$, while the slab variance σ_1^2 remains in a bounded range. This motivates the calibration

scheme

$$\xi_0 = c \sqrt{n \log p}, \quad c \in C,$$

for a finite grid C of positive constants, together with a finite grid

$$\sigma_1 \in \mathcal{S} \subset [\underline{\sigma}, \overline{\sigma}],$$

where $0 < \underline{\sigma} < \overline{\sigma} < \infty$ are fixed. Thus, the theoretical analysis does not determine a unique pair (ξ_0, σ_1) , but restricts calibration to a theoretically admissible low-dimensional family. For each $Q \in \mathcal{Q} = \{Q_{\min}, \dots, Q_{\max}\}$, for a fixed nominal level α , and for each pair $(\xi_0, \sigma_1) \in C \times \mathcal{S}$, we run the full graph inference procedure and obtain a graph estimate $\widehat{A}_\alpha^{(Q, \xi_0, \sigma_1)}$. These candidate graphs are then compared through a Bayesian information criterion,

$$\text{BIC}(\xi_0, \sigma_1; Q, \alpha) = -2 \widetilde{\ell}\left(X; \widehat{A}_\alpha^{(Q, \xi_0, \sigma_1)}\right) + \log n \sum_{i < j} \widehat{A}_{\alpha, ij}^{(Q, \xi_0, \sigma_1)}, \quad (25)$$

where $\widetilde{\ell}(X; A)$ denotes the Gaussian pseudo-loglikelihood associated with the candidate adjacency matrix A . The second term penalizes model complexity through the number of selected edges. For fixed Q , the selected hyperparameters are

$$(\widetilde{\xi}_0^{(Q)}, \widetilde{\sigma}_1^{(Q)}) \in \arg \min_{(\xi_0, \sigma_1) \in C \times \mathcal{S}} \text{BIC}(\xi_0, \sigma_1; Q, \alpha).$$

Then, for the selection of the number of Q , we define the criterion

$$\text{BIC}_Q = -2 \widetilde{\ell}\left(X; \widehat{A}_\alpha^{(Q, \widetilde{\xi}_0^{(Q)}, \widetilde{\sigma}_1^{(Q)})}\right) + (Q - 1) \log p + \frac{Q(Q + 1)}{2} \log\left(\frac{p(p - 1)}{2}\right), \quad (26)$$

where the penalty accounts for the complexity of the block model: the block proportions $\pi = (\pi_1, \dots, \pi_Q)$ contribute $Q - 1$ free parameters, and the symmetric connectivity matrix $\omega \in (0, 1)^{Q \times Q}$ contributes $Q(Q + 1)/2$ distinct entries. Finally, the selected number of blocks is

$$\widehat{Q} \in \arg \min_{Q \in \mathcal{Q}} \text{BIC}_Q.$$

The complete calibration and model selection procedure is summarized in Algorithm ?? in the Appendix ??.

4 Theoretical results

4.1 Error bounds for the adaptive elastic-net estimator

We study the statistical properties of the nodewise adaptive elastic-net estimator used in the M-step of the VEM algorithm. At each iteration, the variational parameters τ and ρ are fixed from the preceding

E-step, so that the quantities

$$p_{ij} = \sum_{q,\ell} \tau_{iq} \tau_{j\ell} \rho_{q\ell}^{i,j}$$

are treated as deterministic. The analysis in this section is therefore conditional on these weights.

Fix a node $i \in \{1, \dots, p\}$ and consider the Gaussian nodewise regression model

$$\mathbf{X}_i = \mathbf{X}_{-i} \beta_i^\star + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_i^2 I_n), \quad (27)$$

where $\beta_i^\star \in \mathbb{R}^{p-1}$ denotes the true regression vector, and K_{ii} is treated as known and fixed. Let

$$S_i = \{j \neq i : \beta_{ij}^\star \neq 0\}, \quad s_i = |S_i|,$$

denote the support of $\beta_i^\star$ and its cardinality. The estimator $\widehat{\beta}_i$ is defined by the nodewise adaptive elastic-net problem (13), with weights given in (14). We impose the following assumptions.

Assumption 4.1 (Sub-Gaussian design). For each node i , the rows of $\mathbf{X}_{-i} \in \mathbb{R}^{n \times (p-1)}$ are independent copies of a centered random vector in $\mathbb{R}^{p-1}$ with covariance matrix $\Sigma_{-i,-i}$. Moreover, there exists a constant $\sigma_X > 0$, uniform in n and p , such that for every unit vector $v \in \mathbb{R}^{p-1}$,

$$\mathbb{E} \left[\exp(t \langle \mathbf{X}_{-i}^{(1)}, v \rangle) \right] \leq \exp \left(\frac{t^2 \sigma_X^2}{2} \right), \quad \forall t \in \mathbb{R},$$

where $\mathbf{X}_{-i}^{(1)}$ denotes a generic row of $\mathbf{X}_{-i}$. In addition, the eigenvalues of $\Sigma_{-i,-i}$ are uniformly bounded:

$$0 < \underline{\kappa} \leq \lambda_{\min}(\Sigma_{-i,-i}) \leq \lambda_{\max}(\Sigma_{-i,-i}) \leq \bar{\kappa} < \infty.$$

Assumption 4.2 (Sparsity and bounded signal). For each node i , the true coefficient vector $\beta_i^\star$ is sparse with support S_i , and satisfies

$$\max_{j \in S_i} |\beta_{ij}^\star| \leq B$$

for some constant $B > 0$, uniform in n and p .

Assumption 4.3 (Separation of adaptive weights). Define

$$\bar{p}_i = \max_{j \in S_i^c} p_{ij}, \quad \underline{p}_i = \min_{j \in S_i} p_{ij}.$$

Assume that $\bar{p}_i < 1/2$.

Assumption 4.4 (Compatibility condition). Let

$$L_i = \frac{5 - 2\underline{p}_i}{1 - 2\bar{p}_i}.$$

There exists a constant $\phi_i > 0$, uniform in n and p , such that for every $\delta \in \mathbb{R}^{p-1}$ satisfying $\|\delta_{S_i^c}\|_1 \leq$

$$L_i \|\delta_{S_i}\|_1,$$

$$\|\mathbf{X}_{-i}\delta\|_2^2 \geq \frac{n\phi_i^2}{s_i} \|\delta_{S_i}\|_1^2. \quad (28)$$

Assumption 4.1 is a standard regularity condition on the design matrix and is automatically satisfied in the Gaussian graphical model considered here. Assumption 4.2 is the usual sparsity and bounded-signal condition. Assumption 4.3 ensures that inactive coordinates receive stronger ℓ_1 -penalization than active ones. Assumption 4.4 is a compatibility condition adapted to the cone constraint induced by the adaptive elastic-net; see, for instance, Meinshausen and Bühlmann (2006); Rothman et al. (2008); Ravikumar et al. (2011); Cai et al. (2011); Loh and Wainwright (2011) for related assumptions in the analysis of ℓ_1 -regularized estimators.

Recall that ξ_0 controls the concentration of the Laplace spike around zero in the prior. The next result shows that taking ξ_0 of order $\sqrt{n \log p}$ yields the usual high-dimensional ℓ_1 -error rate.

Theorem 4.5 (Non-asymptotic ℓ_1 -error bound). Under Assumptions 4.1–4.4, conditionally on the weights (p_{ij}) fixed from the preceding E-step, and provided that

$$\xi_0 \geq C\sqrt{n \log p} \quad \text{and} \quad \xi_0 \geq \frac{BK_{ii}}{\sigma_1^2},$$

for a sufficiently large constant $C > 0$, the nodewise adaptive elastic-net estimator satisfies, with probability at least $1 - 2p^{-c}$,

$$\|\widehat{\beta}_i - \beta_i^\star\|_1 \leq C_i \frac{s_i \xi_0}{n},$$

where $C_i > 0$ depends only on ϕ_i , $\underline{p}_i$, and $\bar{p}_i$.

Under the calibration $\xi_0 \asymp \sqrt{n \log p}$, Theorem 4.5 yields

$$\|\widehat{\beta}_i - \beta_i^\star\|_1 = O_{\mathbb{P}}\left(s_i \sqrt{\frac{\log p}{n}}\right).$$

The proof is given in Appendix ???. The lower bound $\xi_0 \geq C\sqrt{n \log p}$ comes from the concentration of the stochastic term $n^{-1}\mathbf{X}_{-i}^\top \varepsilon_i$. This is the standard minimax-optimal ℓ_1 -rate for sparse linear regression in high dimension (Raskutti et al., 2009). The adaptive weights do not alter the rate itself, but affect the constant through $\underline{p}_i$ and $\bar{p}_i$: better separation between active and inactive weights leads to a tighter bound. Thus, the role of the adaptive elastic-net is not to improve the estimation rate, but to concentrate regularization on the most relevant coordinates.

5 Numerical experiments

We assess the empirical performance of the proposed procedure on synthetic data and on a breast cancer gene expression dataset.

5.1 Synthetic data

We assess the performance of the proposed HSS-GGM procedure on synthetic datasets for which the underlying graph structure is known. Each dataset consists of n independent observations drawn from a multivariate Gaussian distribution with mean zero and precision matrix K_0 , whose sparsity pattern is encoded by a binary adjacency matrix A_0 . To cover a range of realistic dependence structures, we consider four graph topologies, illustrated in Figure 1: a three-hub graph, a stochastic block model graph, a scale-free graph and a band graph.

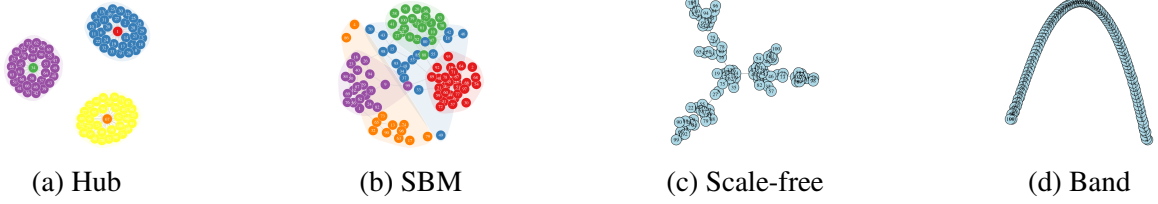


Figure 1: Illustration of the four graph structures considered in the simulation study.

The three-hub graph contains three highly connected hub nodes linked to many peripheral nodes. The stochastic block model graph has higher connection probabilities within groups than between groups. The scale-free graph has a heavy-tailed degree distribution, with a few highly connected nodes and many nodes of low degree. The band graph has a banded adjacency matrix and is representative of ordered dependence structures such as autoregressive models.

For each topology, we follow Kilian et al. (2024) and the `huge` package to generate a positive-definite precision matrix K_0 from the binary adjacency matrix A_0 . Specifically, we define

$$K_0 = \gamma A_0 + (|\lambda_{\min}(\gamma A_0)| + \beta) I_p,$$

with $\gamma = 0.3$ and $\beta = 0.2$. The corresponding covariance matrix is $\Sigma_0 = K_0^{-1}$, from which we draw n independent observations from $\mathcal{N}(0, \Sigma_0)$.

In the main text we consider the high-dimensional setting $n = p = 100$. For each topology, we generate 50 independent datasets, and all results are aggregated across replications.

We compare HSS-GGM with six competing methods: Meinshausen–Bühlmann neighbourhood selection (Meinshausen and Bühlmann, 2006) and graphical lasso (Friedman et al., 2008), both implemented through the `huge` package (Zhao et al., 2012); SIMoNe (Ambroise et al., 2009); GFC-L (Liu, 2013) through the `SILGGM` package (Zhang et al., 2018); BAGUS (Gan et al., 2019); and NSBM-GGM (Kilian et al., 2024), implemented in `noisysbmGGM`.

Graph recovery is evaluated by comparing the estimated adjacency matrix $\hat{A}$ with the true adjacency matrix A_0 . We report the false discovery proportion,

$$\text{FDP} = \frac{|\{(i, j) : \hat{A}_{ij} = 1, A_{0,ij} = 0\}|}{|\{(i, j) : \hat{A}_{ij} = 1\}| \vee 1},$$

and the true discovery proportion,

$$\text{TDP} = \frac{|\{(i, j) : \hat{A}_{ij} = 1, A_{0,ij} = 1\}|}{|\{(i, j) : A_{0,ij} = 1\}|}.$$

For procedures explicitly designed to control the false discovery rate, namely HSS-GGM, GFC-L and NSBM-GGM, we set the nominal level to $\alpha = 0.1$. For BAGUS, we follow Gan et al. (2019) and recover the graph by thresholding posterior edge probabilities at $1/2$. For graphical lasso and Meinshausen–Bühlmann neighbourhood selection, we use the graph selected by the huge implementation. For SIMoNe, the built-in AIC or BIC often selects empty or nearly empty graphs in our experiments. We therefore retain along the regularization path a sparse graph containing about 3% of the possible edges. This level was chosen because it is of the same order as that returned by the other procedures in the settings considered.

Figure 2 summarizes the empirical FDP and TDP. We first consider false discovery control, which is the primary objective of the proposed procedure. In all settings, the empirical FDP of HSS-GGM remains below the nominal level $\alpha = 0.1$. By contrast, several competing procedures exceed the target level in topologies with marked degree heterogeneity, especially hub and scale-free graphs. Although most of these methods are not designed to control the false discovery rate, the observed inflation illustrates the risk of false positives when they are used for graph inference. Even among procedures explicitly targeting false discovery rate control, namely GFC-L and NSBM-GGM, the empirical FDP occasionally exceeds the nominal level in these more difficult settings.

We next consider statistical power. In structured settings such as stochastic block model and hub graphs, and especially when group memberships are known, HSS-GGM achieves the largest TDP, including relative to methods that do not aim to control the false discovery rate and tend to overselect edges. This illustrates the benefit of incorporating group structure into the shrinkage mechanism. Across all graph topologies, HSS-GGM achieves competitive or superior power while maintaining reliable false discovery control. In hub graphs this gain is particularly marked. Under structural misspecification, such as in band graphs, HSS-GGM remains competitive and does not show substantial performance loss, indicating robustness beyond the assumed block structure. Additional simulation results are reported in Appendix ??.

5.2 Breast cancer gene expression network

We apply HSS-GGM to the breast cancer gene expression dataset of Hess et al. (2006), which was also used in the development of SIMoNe (Ambroise et al., 2009). The dataset contains expression profiles from 133 patients with stage I–III breast cancer treated with neoadjuvant chemotherapy. Clinical response is classified as either pathologic complete response (pCR; 34 patients) or residual disease (not-pCR; 99 patients). Following Hess et al. (2006) and Natowicz et al. (2008), who identified a 26-gene signature with strong predictive value for treatment outcome, we restrict attention to this curated gene set, so that $p = 26$.

Because gene expression profiles may differ substantially between responders and non-responders, the assumption of identically distributed observations is unlikely to hold across the full cohort. We

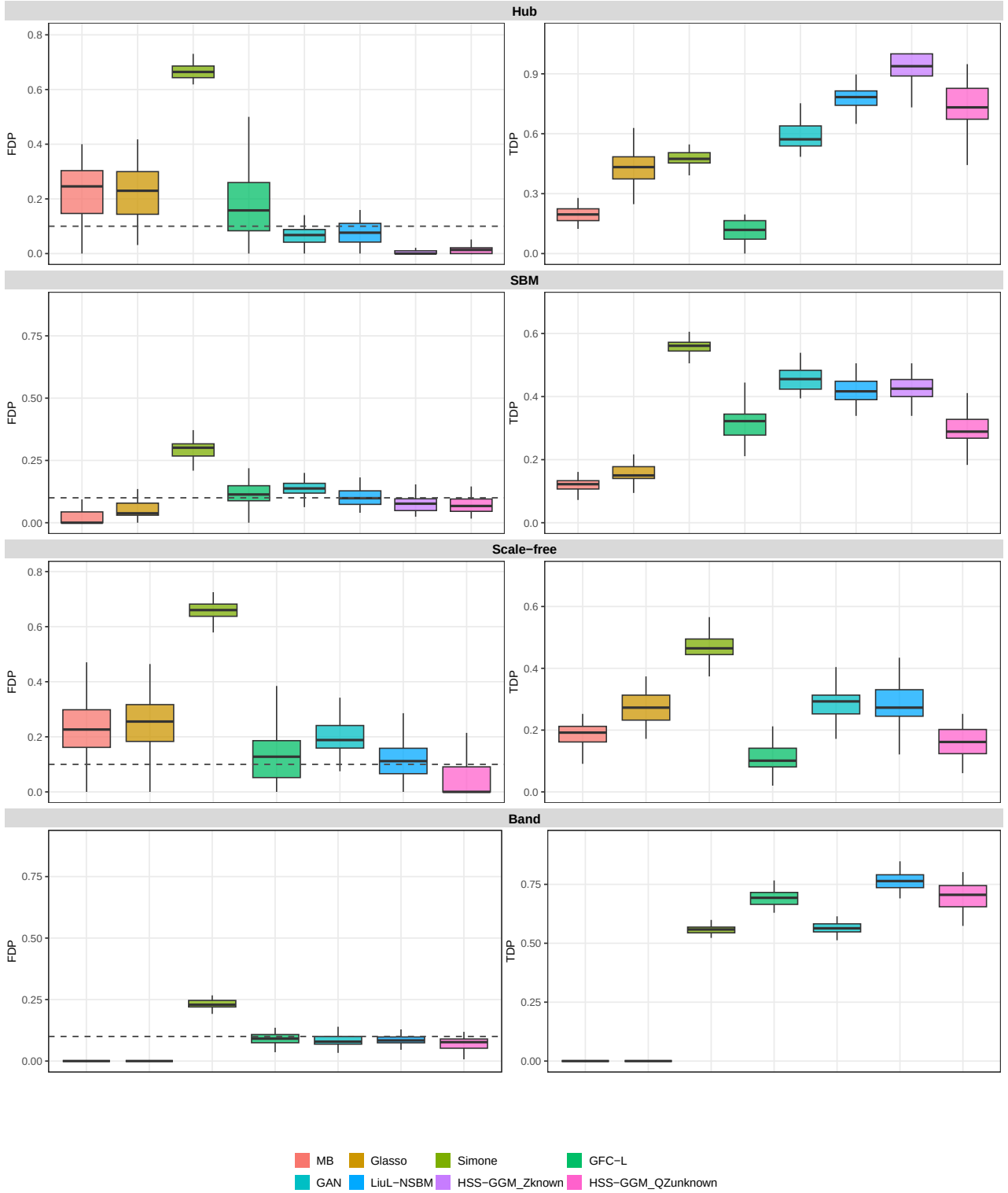


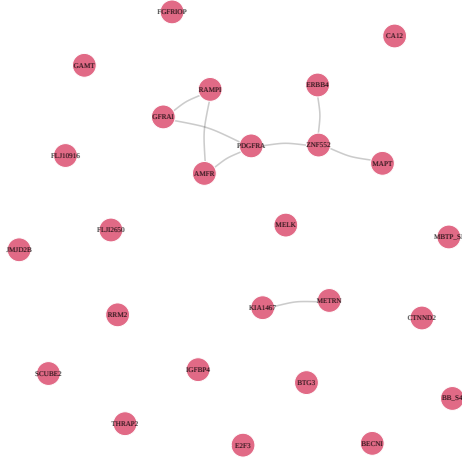
Figure 2: Empirical FDP and TDP distributions for the four graph topologies when $n = p = 100$. For each method, boxplots summarize performance over 50 replications. The horizontal dashed line indicates the nominal false discovery rate level $\alpha = 0.1$.

therefore focus on the pCR subgroup and estimate the corresponding conditional independence network using HSS-GGM at nominal level $\alpha = 0.05$.

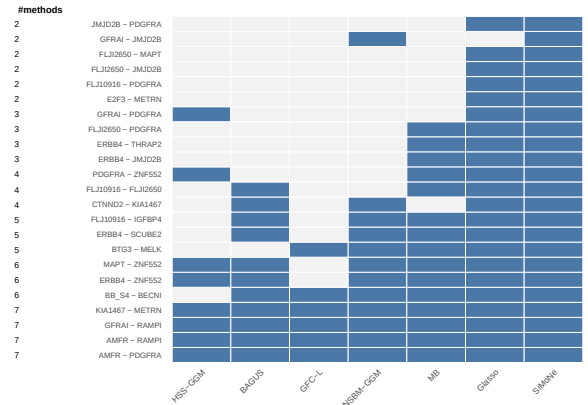
To compare the inferred graph with those obtained by competing procedures, we consider the same methods as in the simulation study. For procedures explicitly designed to control the false discovery rate, namely HSS-GGM, GFC-L and NSBM-GGM, we use the nominal level $\alpha = 0.05$. For BAGUS, we recover the graph by thresholding posterior edge probabilities at $1/2$, following Gan et al. (2019). For graphical lasso and Meinshausen–Bühlmann neighbourhood selection, we use the graph selected by the huge implementation. For SIMoNe, the built-in AIC or BIC may lead to empty or nearly empty graphs; we therefore retain along the regularization path a sparse graph with edge density comparable to that returned by the other procedures.

Figure 3 displays the conditional independence network inferred by HSS-GGM for the pCR subgroup, together with a summary of edge detection across methods. HSS-GGM identifies 8 conditional dependencies among the 26 genes. Each of these edges is also recovered by at least one competing method, indicating substantial agreement across procedures on the most stable interactions.

The inferred network is relatively sparse and overlaps substantially with the graphs obtained by the alternative methods. This is consistent with the simulation results, where HSS-GGM achieved reliable false discovery control while retaining competitive power. In the present data analysis, the method similarly identifies a parsimonious set of conditional dependencies while avoiding the markedly denser graphs produced by some methods that do not explicitly control false discoveries.



(a) Network inferred by HSS-GGM for the pCR subgroup



(b) Edge detection across methods

Figure 3: (a) Conditional independence network inferred by HSS-GGM at nominal level $\alpha = 0.05$ for the pCR subgroup. (b) Comparison of edges detected by different network inference methods. The left column reports the number of methods detecting each interaction.

5.3 S&P 100 stock returns

We next illustrate the method on daily stock returns from the S&P 100 index. We selected $p = 90$ stocks from three sectors, Technology, Healthcare and Financials, with balanced representation

across sectors. Daily adjusted closing prices were collected for the period from June 1, 2024 to December 1, 2024, yielding $n = 126$ observations after preprocessing. Log-returns were computed and standardized to have mean zero and variance one. Because stock returns often exhibit heavy tails, we apply the nonparanormal transformation of Liu et al. (2009) before estimating the graphical model. Additional details on data construction, descriptive statistics and assessment of Gaussianity are given in Appendix ??.

In this setting, a non-zero entry K_{ij} in the precision matrix indicates that the returns of stocks i and j remain associated after conditioning on all other assets. The resulting graph is therefore a partial correlation network, in which edges represent direct conditional dependencies rather than marginal correlations induced by common market or sectoral factors. Such networks are commonly used in financial econometrics to identify groups of assets with related risk profiles and to study channels of risk transmission (Barigozzi and Brownlees, 2019; Giudici et al., 2020).

In this application, sector labels are incorporated as group information in the HSS-GGM prior. This reflects the well-known tendency of firms within the same sector to be exposed to similar economic drivers and therefore provides a natural source of prior structure for the estimation of the dependence network.

Figure 4 displays the network inferred by HSS-GGM at nominal level $\alpha = 0.05$. Because the graph is substantially larger than in the gene expression application, visual comparison with multiple competing methods is less informative, so we focus here on the network produced by HSS-GGM.

The inferred graph shows a clear sectoral organization. Most edges occur within sectors, particularly in Technology and Financials, reflecting shared economic factors within industries. Cross-sector edges are less frequent and tend to involve stocks with relatively high connectivity. Several large-cap firms appear as more central nodes, including MSFT, AVGO and META in Technology, BLK and WFC in Financials, and IDXX and JNJ in Healthcare. These companies are major market participants whose returns often reflect broader sectoral or market-wide information.

Overall, the inferred network is sparse while retaining economically interpretable sector-based structure. This empirical pattern is in line with the simulation results, where HSS-GGM achieved a favourable balance between false discovery control and detection power.

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Group
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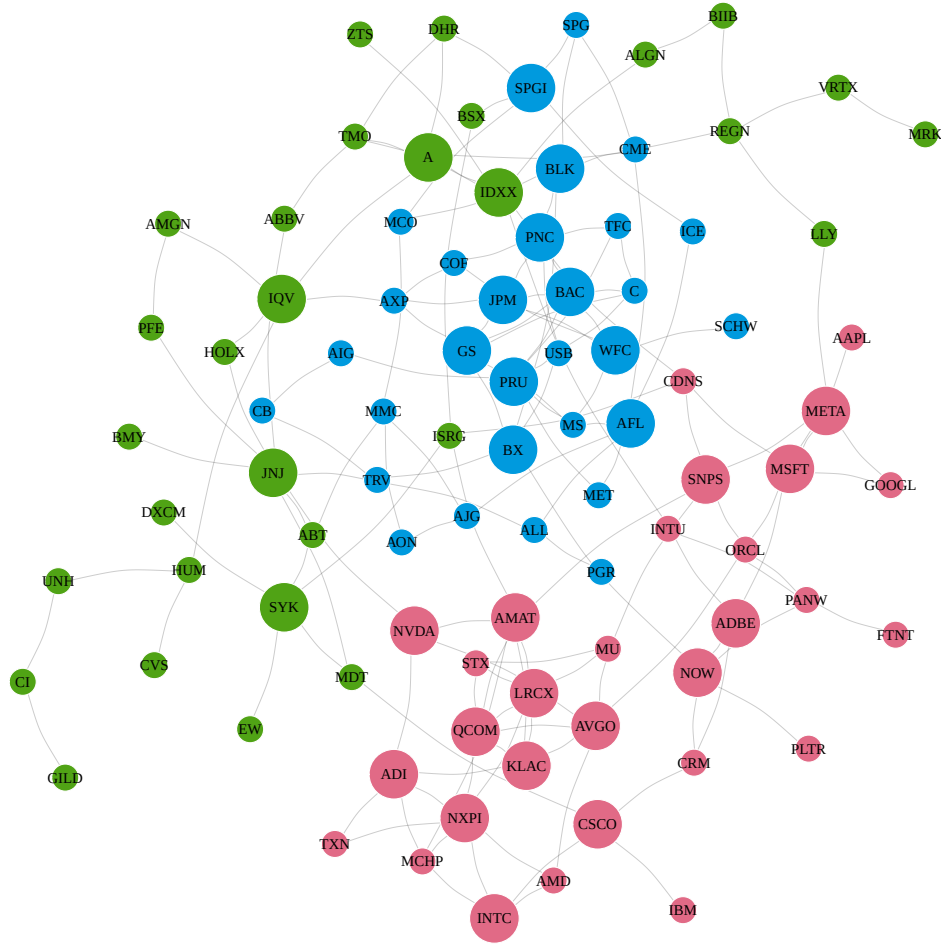


Figure 4: Partial correlation network of S&P 100 stocks inferred by HSS-GGM at nominal level $\alpha = 0.05$. Node colours indicate sector membership: red for Technology, green for Healthcare, and blue for Financials. Edges represent conditional dependencies between stock returns after adjustment for all other assets.

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